

Ajay Rajendra Kumar

rajendrakumar.a@northeastern.edu | [ajaystar8.github.io](https://github.com/ajaystar8) | linkedin.com/in/ajayr-kumar | github.com/ajaystar8

TECHNICAL SKILLS

Programming: Python, C/C++, Java, SQL

Frameworks & Tools: PyTorch, vLLM, Hugging Face, OpenCV, Git

Research Areas: Vision-Language Models, Video Understanding, Multimodal Reasoning, Dense Visual SLAM, Deep Learning

EXPERIENCE

Research Assistant Nov. 2025 - Present

Khoury College of Computer Sciences, Northeastern University (Advisor: Prof. Lorenzo Torresani) Boston, MA

- Improving spatiotemporal reasoning in vision-language models for video question answering, by augmenting frame-level visual features with temporal grounding cues.
- Designing agentic visual grounding pipelines that route VQA queries to expert tools (detection, depth, tracking), targeting higher accuracy on long-form video benchmarks.

Research Assistant Apr. 2025 - Mar. 2026

Visual Intelligence Lab, Northeastern University (Advisor: Prof. Huaizu Jiang) Boston, MA

- Built NeuSLAM, a dense visual-SLAM system that runs real-time on edge devices, by combining neural scene representations with efficient pose tracking; co-authored paper under review at IEEE IROS 2026.
- Demonstrated the system on resource-constrained robot platforms using Jetson Orin Nano; accepted at CVPR 2026 Demo Track and ICRA 2026 S2S Workshop for harsh-domain perception.

Graduate Research Assistant Feb. 2025 - May 2026

D'Amore-McKim School of Business, Northeastern University (Profs. Anna Lamin, Valentina Marano) Boston, MA

- Built ETL pipelines on Corporate Social Responsibility (CSR) spending across 100K+ Indian enterprises over 5 years, isolating firm- and sector-level patterns linking corporate spend to measurable social-impact outcomes; manuscript in preparation.
- Collaborated with IIM Bangalore on a city-scale study quantifying how corporate growth in Bengaluru drives housing affordability shifts.

Software Development Intern Jun. 2023 - Aug. 2023

Fidelity Investments Bengaluru, India

- Cut dashboard query latency from 2+ hours to under 5 minutes on the internal change-ticket management system, by migrating backing store from Oracle to Snowflake and re-architecting query paths against the new warehouse.
- Rebuilt the ticket-management front- and back-end in Angular and Spring Boot, shipping a redesigned workflow used by internal engineering stakeholders to track change requests across the firm.

SELECTED PROJECTS

- Pose Graph Sparsification via Algebraic Connectivity** | *Mobile Robotics, Mar. 2025* – Iterative algebraic connectivity maximization for SLAM back-ends; produced sparser pose graphs with minimal accuracy loss on GTSAM and SE-Sync benchmarks.
- Hallucination Detection in Large Language Models** | *NLP, Mar. 2025* – Reproduced AAAI 2026 hallucination-detection methods across 3+ open-source LLMs; surfaced generalization gaps beyond originally evaluated models.
- Vaccine Misinformation Detection on Social Media** | *NLP, Mar. 2024* – Fine-tuned transformer classifiers and built a similarity-based anomalous-user detector for misinformation on social platforms.

PUBLICATIONS

“**NeuSLAM: Dense Visual SLAM on Edge Devices**” – under review at *IEEE IROS 2026*; accepted to *ICRA 2026 S2S Workshop* and *CVPR 2026 Demo Track* (co-author)

“**Addressing Vaccine Misinformation on Social Media by Leveraging Transformers and User Association Dynamics**” – *ICMLDE 2023* (co-first author)

HONORS & AWARDS

- 2025** – Most Innovative Project (among 40+ submissions), PhD-level Deep Learning Course (CS7150), Northeastern University
- 2023** – Undergraduate Sponsorship Grant, Manipal Academy of Higher Education (MAHE)
- 2022** – National 2nd Runner-Up, Daimler India Commercial Vehicles Hackathon
- 2022** – 1st Runner-Up, Transfusion Medicine Hackathon
- 2021** – Winner, Space Odyssey Web Development Hackathon
- 2021** – Scholar’s Scholarship Award, Manipal Academy of Higher Education (MAHE)

EDUCATION

Northeastern University Boston, USA

M.S. in Computer Science; GPA: 4.0/4.0 Sep. 2024 - May 2026

Manipal Institute of Technology Manipal, India

B.Tech. in Computer Science (Minor: Computational Intelligence); GPA: 9.3/10.0 Aug. 2020 - Jul. 2024